The due date for submitting this assignment has passed.
Due on 2026-03-25, 23:59 IST.

You may submit any number of times before the due date. The final submission will be considered for grading.

You have last submitted on: 2026-03-25, 07:24 IST

Note: This assignment will be evaluated after the deadline passes. You will get your score 48 hrs after the deadline. Until then the score will be shown as Zero.

Multiple Select Questions (MSQ):

1) A function $T : V \to W$ between two vector spaces V and W is said to be a linear transformation if the following conditions hold: ***1 point***

Condition 1: $T(v_1 + v_2) = T(v_1) + T(v_2)$ for all $v_1, v_2 \in V$.
Condition 2: $T(cv) = cT(v)$ for all $v \in V$ and $c \in \mathrm{R}$.

Consider the following function:

$$T : \mathrm{R}^2 \to \mathrm{R}$$
$$T(x, y) = \begin{cases} 3x & \text{if } y = 0 \\ 4y & \text{if } y = 0 \end{cases}$$

Which of the following statements is true?

- [ ] Condition 1 holds.
- [x] Condition 1 does not hold.
- [x] Condition 2 holds.
- [ ] Condition 2 does not hold.

Yes, the answer is correct.
Score: 1

Accepted Answers:

Condition 1 does not hold.
Condition 2 holds.